

Customer Shopping Behavior Analysis

Business Problem Statement & Analytical Approach

Project Objective

Analyze retail customer shopping data to identify purchasing patterns, customer behavior, product and category performance, and factors that may influence spending and purchase activity.

1. Business Context

Retail businesses collect customer-level transaction and behavioral data across products, categories, locations, seasons, payment methods, promotions, shipping preferences, subscriptions, and purchase frequency. However, raw transactional data does not directly provide decision-ready insights.

This project converts the raw customer shopping dataset into a structured analytical solution that can help stakeholders understand customer purchasing behavior and identify patterns that are relevant to retail decision-making.

2. Business Problem

The primary business problem is to understand **how customers purchase, what they purchase, how much they spend, and how customer characteristics and purchasing conditions relate to shopping behavior.**

The analysis will examine differences across customer demographics, product categories, locations, seasons, subscriptions, discounts, promotions, payment methods, shipping types, and purchase frequency.

3. Dataset Overview

Attribute	Value
Rows	3,900 customer purchase records
Columns	18 attributes
Missing values	37 missing values in <i>Review Rating</i>
Duplicate rows	0 duplicate rows identified
Main measure	Purchase Amount (USD)
Key dimensions	Age, Gender, Item, Category, Location, Size, Color, Season, Subscription, Shipping, Payment, Frequency

4. Key Business Questions

- What are the overall customer purchasing and spending patterns?
- Which product categories and individual items generate the highest purchase activity or spending?
- How does purchase behavior vary by gender, age group, location, and season?
- How do subscription status and promotional discounts relate to purchasing behavior?
- How does purchase frequency vary across different customer groups?

- Which payment methods and shipping types are most commonly used?
- How do discounts and promo-code usage relate to purchase amounts?
- What patterns can be identified from customer review ratings?
- Which customer segments show distinct purchasing characteristics?
- What actionable insights can be derived for retail decision-making?

5. Analytical Approach

The project follows a structured end-to-end analytics workflow. Each stage produces an output that is used in the next stage.

Stage	Tools	Key Activities	Output
1. Business Understanding	—	Define the business problem, objectives, stakeholders, scope, and key business questions.	Problem statement & analytical requirements
2. Data Loading & Profiling	Python, Pandas	Load the CSV, inspect dimensions, data types, columns, summary statistics, missing values, and duplicates.	Initial data-quality assessment
3. Data Cleaning	Python, Pandas	Handle missing <i>Review Rating</i> values using a justified imputation method, verify data types, check duplicates, and validate categorical/numerical fields.	Cleaned dataset
4. Feature Engineering	Python, Pandas	Create analysis-ready fields such as age groups and other derived variables required for business analysis.	Analysis-ready dataset
5. Exploratory Data Analysis	Python, Pandas, Matplotlib/Seaborn	Analyze distributions, trends, category performance, customer segments, spending patterns, and relationships between variables.	Exploratory findings & visualizations
6. SQL Business Analysis	PostgreSQL / SQL	Query the cleaned data using aggregations, GROUP BY, JOINs where applicable, subqueries, CASE statements, CTEs, and window functions.	Business metrics & SQL insights
7. Power BI Dashboard	Power BI, DAX	Build KPI cards, charts, slicers, and interactive views for customer, product, category, spending, and behavioral analysis.	Interactive dashboard
8. Insight & Recommendation Development	Python, SQL, Power BI	Translate validated findings into clear business observations and practical areas for stakeholder consideration.	Decision-ready insights
9. Documentation & Portfolio	GitHub, Markdown, PDF/PPT	Document methodology, assumptions, data-quality decisions, SQL analysis, dashboard, findings, and project limitations.	Complete portfolio project

Important data-quality decision: The missing-value treatment for *Review Rating* should be selected after comparing the overall and category-level distributions (for example, mean, median, spread, and potential outliers). The chosen method should be documented in the Python notebook rather than copied without validation.

6. Expected Deliverables

- **Python Notebook:** data loading, profiling, cleaning, feature engineering, EDA, and visual analysis.
- **Cleaned Dataset:** validated CSV ready for downstream analysis.

- **SQL Analysis:** business-focused SQL queries and derived metrics.
- **Power BI Dashboard:** interactive dashboard with KPIs, filters, and visual insights.
- **Business Report:** methodology, findings, data-quality decisions, and business implications.
- **GitHub Repository:** organized project files, documentation, and reproducible workflow.

7. Expected Business Value

The analysis is intended to help a retail stakeholder understand customer behavior through measurable patterns in spending, product/category activity, customer segments, promotions, subscriptions, and purchasing conditions. The project focuses on evidence from the dataset rather than unsupported assumptions.

8. Scope and Limitations

- The analysis is based only on the fields available in the provided dataset.
- The dataset does not establish causal relationships between variables; observed relationships should be interpreted as associations.
- The data period and business context are not provided in the source dataset, so time-based conclusions should be limited to the available information.
- Customer-level identifiers should be treated as analytical identifiers and not used to expose personally identifiable information.
- Recommendations should be based on validated analytical findings and business context.

9. One-Line Problem Statement

“Analyze customer shopping behavior to identify actionable patterns in spending, products, categories, customer segments, promotions, subscriptions, and purchasing preferences, and present the findings through Python, SQL, and Power BI.”

Prepared for portfolio and GitHub documentation • Dataset: customer_shopping_behavior.csv